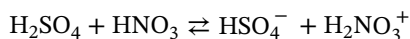


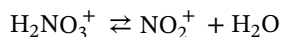
The overall reaction for the generation of an NO_2^+ electrophile:



can be stated in two separate, reversible steps as a protonation followed by a decomposition (Reactions 1 and 2).



Reaction 1



Reaction 2

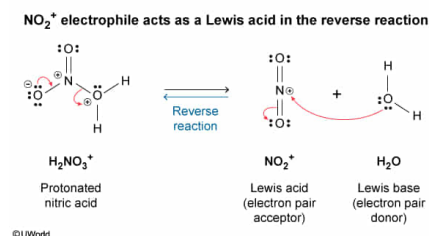
Considering both the forward and reverse directions of these two reactions, which molecule acts as a Lewis acid, but NOT as a Brønsted-Lowry acid?

- ☐ A. H_2SO_4 [13%]
☐ B. HNO_3 [21%]
☐ C. H_2NO_3^+ [19%]
☒ D. NO_2^+ [45%]

Correct

44% answered correctly

Explanation:



By the [Brønsted-Lowry definitions](#), an acid is a molecule that donates a proton (H^+ ion), and a base is a molecule that accepts a proton. Although the Brønsted-Lowry definitions are sufficient to identify acids and bases in most cases, these definitions are inadequate for molecules that lack **acidic protons** in their structure yet exhibit acid-base behavior. To overcome this limitation, the **Lewis definitions** were developed. Accordingly, a **Lewis acid** is defined as an **electron pair acceptor**, and a **Lewis base** is defined as an **electron pair donor**.

In Reactions 1 and 2, all chemical species except H_2O and NO_2^+ follow the Brønsted-Lowry definition of acid-base behavior. Notably, NO_2^+ lacks H atoms and cannot be classified by the Brønsted-Lowry definition. However, in the reverse reaction of Reaction 2, the NO_2^+ electrophile accepts a pair of electrons donated by H_2O . Therefore, applying the Lewis definitions of acids and bases, H_2O is acting as a Lewis base (electron pair donor) and NO_2^+ is acting as a Lewis acid (electron pair acceptor).

(Choice A) Although the H^+ ion acts as a Lewis acid (electron pair acceptor), H_2SO_4 functions as a Brønsted-Lowry acid by donating H^+ ions.

(Choice B) HNO_3 acts as a Brønsted-Lowry base (H^+ acceptor) and also as a Lewis base (electron pair donor) in Reaction 1, but HNO_3 does not act as a Lewis acid (electron pair acceptor).

(Choice C) H_2NO_3^+ is an H^+ donor (Brønsted-Lowry acid) in the reverse of Reaction 1 and decomposes in Reaction 2.

Educational objective:

A Lewis acid is a molecule that acts as an electron pair acceptor, and a Lewis base is a molecule that acts as an electron pair donor. The Lewis definitions more broadly define acids and bases to include some compounds that lack

acidic protons.

Time spent:0 sec
Subject:
General Chemistry

QID:402375
System:
Interactions of Chemical Substances